

CPSC 4300/6300: Assignment 4

Instructions

- Please submit your solution by 11:59PM November 14.
- Submissions should be made through **Canvas**.
- Please complete homework individually.
- Please submit a **single PDF/Word file** including your solutions and the code of your solutions. Feel free to use any third-party libraries.

1. Neural Networks (30 points):

A neuron uses the ReLU activation function:

$$f(x) = \max(0, x)$$

and computes its pre-activation as:

$$z = w_1x_1 + w_2x_2 + b$$

Given the following input and parameter values:

Case	x_1	x_2	w_1	w_2	b
A	1.0	2.0	0.5	0.3	-0.2
B	1.0	2.0	-0.5	-0.8	-0.5
C	2.0	3.0	-1.0	-1.0	-1.0

- For each case, compute the pre-activation z and output $f(z)$
- Based on these numbers, which neuron appears to have “stopped activating”?
- What is the impact of having “stopped activating” neurons in a network? How to avoid this problem?

2. Dimensionality Reduction & Clustering (70 points):

Note: This question is based on the Most-Recent-US-College-Scorecard.csv data which can be found in Assignment 4 on Canvas

The dataset ([more info](#)) describes almost every postsecondary institution in the United States, reporting measures of cost, selectivity, completion, and post-graduation outcomes. Each row corresponds to a single college or university, and each numeric column summarizes a key institutional characteristic such as:

Variable	Description
ADM_RATE	Admission rate (fraction of applicants admitted)
TUITIONFEE_IN / TUITIONFEE_OUT	Annual tuition (in-state / out-of-state)
C150_4	4-year completion rate
MD_EARN_WNE_P10	Median earnings of former students 10 years after entry
UGDS	Undergraduate enrollment
AVGFACSL	Average faculty salary
PELL_COMP_ORIG_YR6_RT	Pell Grant recipient completion rate

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Only use the features in the above table,

- (a) Plot the PCA *cumulative explained variance* versus the number of components. How many components are needed to reach 90%?
- (b) Create scatter plots for:
- PCA (2 components), color by *TUITIONFEE_IN*
 - Kernel PCA (RBF), color by *C150_4*
 - t-SNE (2 components), color by *MD_EARN_WNE_P10*
 - Compare PCA vs. Kernel PCA and t-SNE
- (c) Apply K-Means and DBSCAN on your PCA-reduced data (2 components).
- Show both cluster scatter plots.
 - Report their Silhouette Scores.

Note: If your code requires a random seed, use 15 to ensure reproducibility.

CPSC 4300/6300: Assignment 3
